IMPERIAL

CCIMA Centre for Climate Impact Adaptation

BCKIC Bhubaneswar City Knowledge Innovation Cluster Foundation

DigiMic

Model-guided microbial solutions for climate-resilient agriculture

Combining microbial ecology, experiments and mechanistic models to guide the design of useful, resilient microbial communities.

THE PROBLEM

A microbial treatment that works in the laboratory may fail in a different soil or season. Its microbes must establish among resident organisms, keep functioning as conditions change, and deliver a measurable benefit to the crop. Selecting strains for a useful trait alone does not resolve these ecological challenges.

THE SOLUTION: DIGIMIC

DigiMic is our developing open modelling toolkit for understanding and designing microbial communities. Mechanistic models represent how microbes use resources, exchange metabolites and respond to their environment. Combined with experiments, they help us ask **which combinations are most likely to persist, what they can do, and under which conditions.**

MEASURE
Microbes, traits and conditions

→

MODEL
Interactions and community dynamics

→

DESIGN
Candidate microbial combinations

→

TEST & REFINE
Function under relevant conditions

AN AGRICULTURAL APPLICATION: RELIABLE BIOFERTILISERS

A current development focus is **DigiMic-P**: using the toolkit to design microbial communities that help wheat access phosphorus already present in soil. The proposed approach combines microbes with a shared useful function but different resource needs and environmental tolerances. As conditions change, different members may sustain the desired function. Experiments will test this principle and identify when additional helper microbes improve establishment.

WHAT THIS COULD ENABLE

- **More focused development:** prioritise candidate communities and informative experiments before committing to larger trials.
- **Resilience by design:** evaluate persistence and function across relevant soil, nutrient and climate conditions, rather than a single ideal setting.
- **Evidence for practical use:** test nutrient delivery and crop performance against controls, and identify the conditions where benefits hold or fail.

PARTNER WITH US

We seek agricultural, industry and public-sector partners to **co-develop and fund a pilot** for a defined crop, soil and climate challenge. Partners can contribute microbial resources, measurements, trial settings and product-development expertise. Funding would connect modelling and controlled experiments to staged validation, with success measures agreed at the outset.